VHGRAD

Development of a Didactic Deep Learning Framework from Scratch

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2026

Technical Progression of the VHgrad Framework

1. Foundations of Neural Computation
Historical context and fundamental concepts underlying modern neural networks.

2. Building the First Components
Single neurons, multi-neuron layers, tensors and matrix operations with NumPy.

3. Structured Layer Design
Encapsulation of dense layers into reusable classes with organized data handling.

4. Non-Linearity through Activation
Step, linear, sigmoid, ReLU and Softmax activation functions and their roles.

5. Measuring Network Error
Cross-entropy loss formulation and accuracy metrics for classification.

6. Parameter Optimization
First principles of gradient-based learning and iterative parameter updates.

7. Calculus of Learning
Numerical and analytical derivatives, partial derivatives and the chain rule.

8. Gradient Flow through the Network
Full backward pass with analytically derived gradients across layers and losses.

9. Advanced Optimization Methods
SGD, learning rate scheduling, momentum and adaptive optimizers including Adam.

10. Controlling Generalization
Weight regularization, dropout and validation strategies to reduce overfitting.

11. Extending to Other Task Types
Binary classification and regression with appropriate outputs and loss functions.

12. Training on Real Data
End-to-end pipeline covering loading, preprocessing, batching and model training.

13. Deployment and Reuse
Parameter serialization, model loading and inference on unseen data.